Introduction

This report aims to provide an analysis of the thermal conditions within a tent and propose designs or solutions for improving thermal comfort for tent occupants that an outdoor equipment supplier could implement. To achieve this, analysis of a typical hot and cold scenario was performed, analysing the key contributors to thermal losses that may cause a tent occupant to feel thermally uncomfortable.

To effectively solve the aim of this report, research into the conditions deemed uncomfortable were initially undertaken to set out the criteria that proposed solutions could be compared to. The problem was limited to the UK and temperate climates were analysed, so any solutions and considerations should take this target audience into account. For analysis, the most popular camping location in the UK [1], the Lake District was used and all values for weather conditions were taken from the Keswick weather station. After the thermal conditions in the hot and cold scenarios were found, they were compared to the comfortability criteria and following this, solutions were proposed to address the key contributors to thermal discomfort. These solutions considered design features and objectives for the target audience to ensure solutions improved thermal comfort without trade-offs in other important design goals. After a brief thermal analysis for the proposed solutions, the most effective design suggestions were highlighted for further research and development by the outdoor equipment supplier.

Defining the Problem

Defining Thermal Comfort

The theory used to define comfort was an energy balance between the heat produced by a tent occupant and the heat taken away from surroundings. The idea was that if the person produces energy at a metabolic rate of 58.2 W/m^2 (1 standard met at rest) then to be in perfect equilibrium (perfect comfort), energy should be taken by surroundings at the same rate. Any additional energy is considered stored energy, S and any net negative energy is assumed to be lost (and the occupant feel cold) and any net positive S suggest energy storage (the occupant fee

- From ASHRAE: $PMV = (0.303e^{-0.036M} + 0.028)L$

  where $M = 58.2$ $PMV = 0.0653L$

± 1 PMV in body storage is considered uncomfortable.

$L = \pm 15.3$ W/m^2 Additional heat storage that would be considered uncomfortable

Modelling Assumptions

Tent Selection and Material Properties

The first consideration for this analysis was the tent design and material properties. The tent selected was a best-selling model from ultimate outdoors [3]; the best-selling model was selected for its popularity and appeal to the largest audience, allowing solutions to be most widely relevant. The Vango Banshee 200 is a two-person tent designed to be lightweight for trekking (an important feature when proposing solutions) with the material properties defined below.

Table 1: Tent Material Properties [4][5]

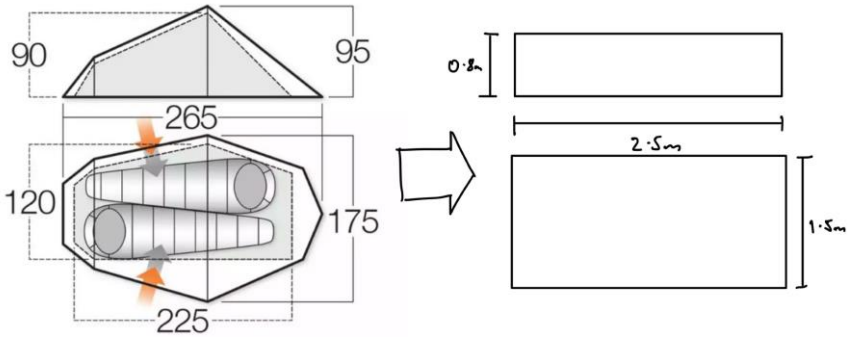
	Material	Emissivity ε	Absorptivity α
Groundsheet	Polyester (6000HH)	0.8	0.35
Inner	Breathable Polyester	n/a	n/a
Outer	Polyester (3000HH)	0.8	0.35

The values from above for emissivity and absorptivity were found from [4] and [5] respectively and would be used for analysis. The inner layer of the tent was ignored due to it being breathable, this would cause all values for convective and radiative transmission within the tent to be an overestimate (assuming no barrier when there was one in reality). Due to

the design requirement for the tent of being lightweight, a thin-walled assumption was made for the tent wall, so conductivity of the tent was ignored but radiative and convective effects were modelled around a boundary.

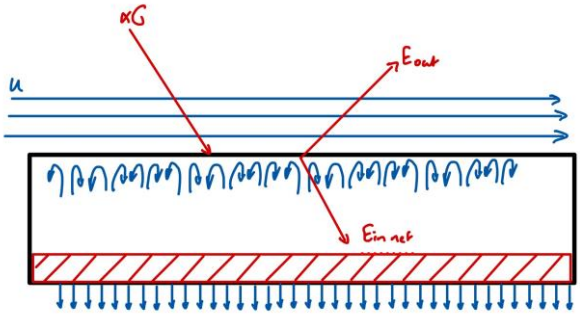
Tent Geometry Simplification

To simplify the tent's dimensions from its complex geometry, it was modelled as a cuboid and further reduced to a series of flat plates so losses through the walls were ignored. The individual effects of this big assumption were discussed in more detail during the analysis of conduction, convection and radiation below. The tent geometry was transformed as seen below.



Occupant Assumptions and Simplifications

Assumptions on the tent occupants were as follows: there were 2 occupants in the tent (the tent is used as intended); human skin temperature is 33 degrees C [6]; tent occupants were assumed to have no additional insulation in the form of clothes or a sleeping bag and the emissivity of human skin was assumed to be 0.98 [7]; and the occupants were modelled to be laying down and in perfect contact with the floor of the tent and occupy the full floor surface area. This allowed another flat plate assumption to be made with a constant surface temperature of 33 degrees C. The systems were also assumed to have be in steady state which is reasonable as the tent has a small mass required very little energy to raise its temperature so any transient heating period would be small. A diagram of the effective system to be modelled is shown below.



These modelling assumptions allowed hot and cold analysis to be performed in 2 parts, the radiative and convective heat losses (dominant on the top surface of the occupants) and the dominant conductive heat losses between the floor and the occupants.

Weather Data

Table 2: Weather and Soil info at peak summer and winter [8][9]

	Peak temp (°C)	Avg. Wind spd. (m/s)	Soil Temp (0.3m) (°C)
January	2	3.5	3.5
July	20	1.7	18.5

Analysis of Hot and Cold Cases

Both typical hot and cold systems were modelled as similar cases, same principal transfers of energy, with different values (which in some cases will be negative indicating a different direction of heat transfer). As previously discussed, the tent was split into 2 systems for analysis (convection and radiation on top, and conduction on the bottom of the occupants). For all further analysis, the weather and soil data was taken from table [2] and material properties for the tent were taken from table [1]. Average Nusselt number were found rather than local values to ensure energy balance was applied over the entire plate.

Convective and Radiative System

Energy Balance

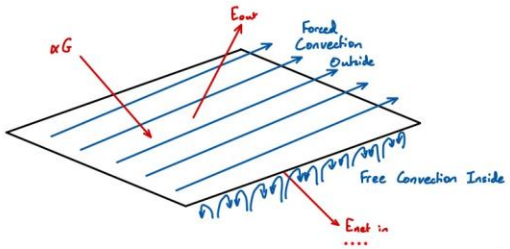
To find the net radiation from the tent surface to the person, an in-depth study was conducted to determine the temperature of the tent surface (that would be the same for the top and bottom surface as the tent is modelled as thin walled). This study applied an energy balance between incoming solar radiation, emitted radiation (both inward and outward) and the heat losses to the surroundings from the wind externally in forced convection and the free convection inward to the tent. For this study, an initial approximation for surface temperature was made and iterated to find the true value for surface temperature.

1. Setting up energy balance. First, the heat fluxe from solar radiation:

$$G = \alpha G + \rho G + \tau G$$

• since the tent is assumed to be thermally opaque, $\tau = 0$

as the energy to/from the tent is the concern, αG can be inspected and the effective system can be seen below



Where the surface is assumed to be thin so T_s is the same on both surfaces. This system can be quantified in the energy balance below:

$$\alpha G = \varepsilon\sigma(T_s^4 - T_{out}^4) + h_{out}(T_s - T_{out}) + E_{net\,in} + h_{in}(T_s - T_{in})$$

Where T_{out} is the ambient air temperature, T_{in} is the internal air temperature and for hot and cold cases:

Hot Case		
T_{out}	= 20°C	[ ]
T_{in}	= 27°C	[ ]
G	= 900 W/m²	[ ]
α	= 0.35	[ ]
ε	= 0.8	[ ]

Cold Case		
T_{out}	= 2°C	[ ]
T_{in}	= 3°C	[ ]
G	= 0 W/m²	[ ]
α	= 0.35	[ ]
ε	= 0.8	[ ]

External Forced Convection

Here, wind was assumed to be acting completely horizontally over the flat plate. As the tent is relatively low (comparative to wind speed with relation to height), the wind was only modelled to act over the top surface of the tent and not on the walls, an assumption that would underestimate the heat loss to the wind. The data for the hot and cold cases was input into the workings below and the air properties at various temperatures were found using interpolation and table A-5 from the textbook and notes. The relevant equations for Reynolds, Prandtl and Nusselt number for each specific case (shown below) were used to find the heat transfer coefficient for the air outside the tent.

2. External Forced Convection

Using an initial approximation for surface temp, for the hot case: 32°C (305K) and the cold case: 0°C (273K) the film temp is:

Hot: $T_{f_1} = \frac{305+293}{2} = 299\,K$ Cold: $T_{f_1} = \frac{273+273}{2} = 274\,K$

Taking values of 300K for hot and interpolating between 250 and 300K for cold from table A-5 (from notes) air properties found to be:

Hot: ↳ $\rho = 1.177\ (kg/m^3)$
$c_p = 1005.7\ (J/kgK)$
$\mu = 1.8462 \times 10^{-5}\ (kg/ms)$
$k = 0.02624\ (W/mK)$
$u = 1.7\ (m/s)$ [ ]
$L = 2.5m$ (from model)

↳ $Re = \frac{\rho u L}{\mu} = 270{,}948$
$Pr = \frac{\mu c_p}{k} = 0.7076$

$Re < 5 \times 10^5$ Laminar, $\bar{Nu}$ correlation:

$\bar{Nu} = 0.664\, Re^{\frac{1}{2}} Pr^{\frac{1}{3}} = 307.99$

$h_{out} = \bar{Nu}\frac{k}{L} = 3.23\ W/m^2K$

Cold: ↳ $\rho = 1.2998\ (kg/m^3)$
$c_p = 1005.5\ (J/kgK)$
$\mu = 1.7177 \times 10^{-5}\ (kg/ms)$
$k = 0.02418\ (W/mK)$
$u = 3.5\ (m/s)$ [ ]
$L = 2.5m$ (from model)

↳ $Re = \frac{\rho u L}{\mu} = 662{,}120$
$Pr = \frac{\mu c_p}{k} = 0.7143$

$Re < 10^7$ Mixed, $\bar{Nu}$ correlation:

$\bar{Nu} = Pr^{\frac{1}{3}}(0.037 Re^{0.8} - 871) = 721.93$

$h_{out} = \bar{Nu}\frac{k}{L} = 6.98\ W/m^2K$

Doing a quick check to ensure these cases are dominated by forced convection:

Hot: $\frac{Gr}{(Re)^2} \approx 9 \times 10^{-4} << 1$ confirming forced convection is dominant.

Cold: $\frac{Gr}{(Re)^2} \approx 1.4 \times 10^{-4} << 1$ confirming forced convection is dominant.

Internal Free Convection

Assuming no ventilation for both cases (no vents on the tent and the door is shut) the internal wind speed was assumed to be 0 for both cases. The same initial approximations for surface temperature were used and from studies [10] and [11] the values of internal air temperature were estimated in the workings below. The Grashof, Prandtl and Nusselt numbers were again used for these cases to find the heat transfer coefficient of air inside the tent.

3. Free Internal Convection

Using an initial approximation for surface temp for the hot case: 32°C (305K) and for the cold case: 0°C (273K), the film temp is:

Hot: $T_{f_1} = \frac{305+300}{2} = 302.5K$ Cold: $T_{f_1} = \frac{273+276}{2} = 274.5K$

Interpolating values on table A-5 (from notes) between 300 and 350K for hot air and 250 and 300K for cold air, air properties found to be:

Hot:
- $\rho = 1.1682\ (kg/m^3)$
- $c_p = 1005.9\ (J/kgK)$
- $\mu = 1.8576 \times 10^{-5}\ (kg/ms)$
- $k = 0.02643\ (W/mK)$
- $\nu = 15.94 \times 10^{-6}\ (m^2/s)$
- $L = \frac{A}{P} = 0.46875\ (m)$
- $\beta = \frac{1}{T_f}$ (ideal gas)

$Gr = \frac{g\beta(T_w - T_\infty)L^3}{\nu^2} = 6.57 \times 10^7$

$Pr = \frac{\mu c_p}{k} = 0.7068$

Lower surface of horizontal heated plate: $C = 0.27$, $m = \frac{1}{4}$

$\bar{N_u} = C(Gr\ Pr)^m = 22.288$

$h_{in} = \bar{N_u}\frac{k}{L} = 1.26\ W/m^2K$

Cold:
- $\rho = 1.2975\ (kg/m^3)$
- $c_p = 1005.5\ (J/kgK)$
- $\mu = 1.7201 \times 10^{-5}\ (kg/ms)$
- $k = 0.02422\ (W/mK)$
- $\nu = 13.46 \times 10^{-6}\ (m^2/s)$
- $L = \frac{A}{P} = 0.46875\ (m)$
- $\beta = \frac{1}{T_f}$ (ideal gas)

$Gr = \frac{g\beta(T_w - T_\infty)L^3}{\nu^2} = 6.10 \times 10^7$

$Pr = \frac{\mu c_p}{k} = 0.7141$

Lower surface of horizontal cooled plate: $C = 0.15$, $m = \frac{1}{3}$

$\bar{N_u} = C(Gr\ Pr)^m = 52.778$

$h_{in} = \bar{N_u}\frac{k}{L} = 2.73\ W/m^2K$

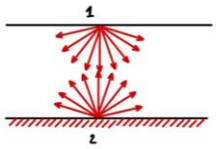
- No forced convection assuming no ventilation. → free convection

Radiation

For all cases of radiation, as the system was modelled as infinite parallel plates with solar radiation being emitted from directly above the tent, so all relevant view factors were assumed to be one on the tent surface and the wall radiation was negated. Externally, all the values were known from tables [1] and [2] so these could be substituted straight into the energy balance equation at the end. The internal radiation for the hot case used the values shown in the workings to show the net radiation from the tent surface to the occupants, however for the cold case, study [11] based the internal temperatures off a well-insulated person (in a sleeping bag) and this model assumed no insulation of the occupants. For this reason, the radiative transfer between the person and the tent wall was neglected for the cold analysis, which would result in a slight underestimate for surface temperature, as in reality not all radiation would be completely insulated.

4. Internal Radiation

An infinite parallel plate assumption was made here, so the view factor



$F_{12} = F_{21} = 1$

where surface 2 is the tent occupies modelled as a flat surface whose $\epsilon_2 = 0.98$ [ ]

So $E_{net\,in} = \epsilon_1\sigma T_s^4 - \epsilon_2\sigma T_2^4$

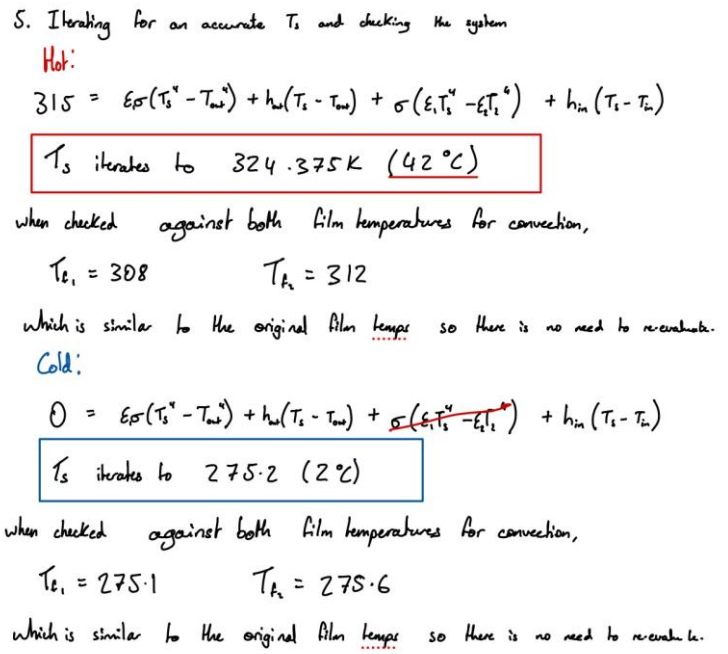
Hot:
- $T_2 = 33°C\ (306K)$ [ ]
- $T_s = 32°C\ (305K)$
- $\epsilon_1 = 0.8$
- $\epsilon_2 = 0.98$ [ ]
- $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$

Cold: Radiation assumed to be negligible

Iterating Surface Temperature

These h values and the radiation information were then substituted into the energy balance equation and iterated until the surface temperatures converged to a balanced equation (when the RHS converged to 315W/m^2 for the hot case and 0 for the cold case (no solar radiation)). These surface temperature values were then used to calculate the new film

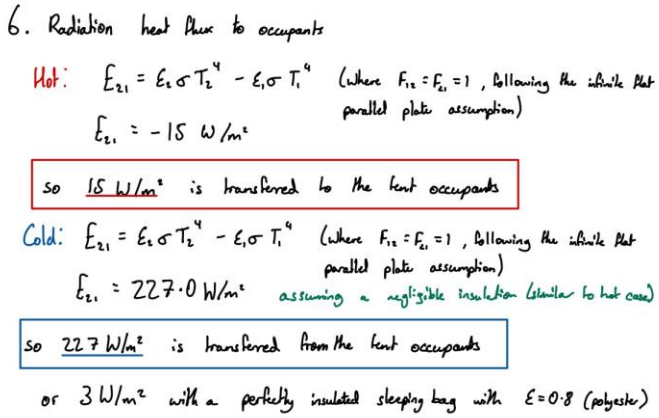
temperatures around the surface at both faces and compared to ensure they were similar. This ensured the correct values for the properties of air were used by ensuring the variation in properties of temperature change did not have a drastic effect on the convective calculations. The numerical calculations are shown below.



Heat Flux to Occupants

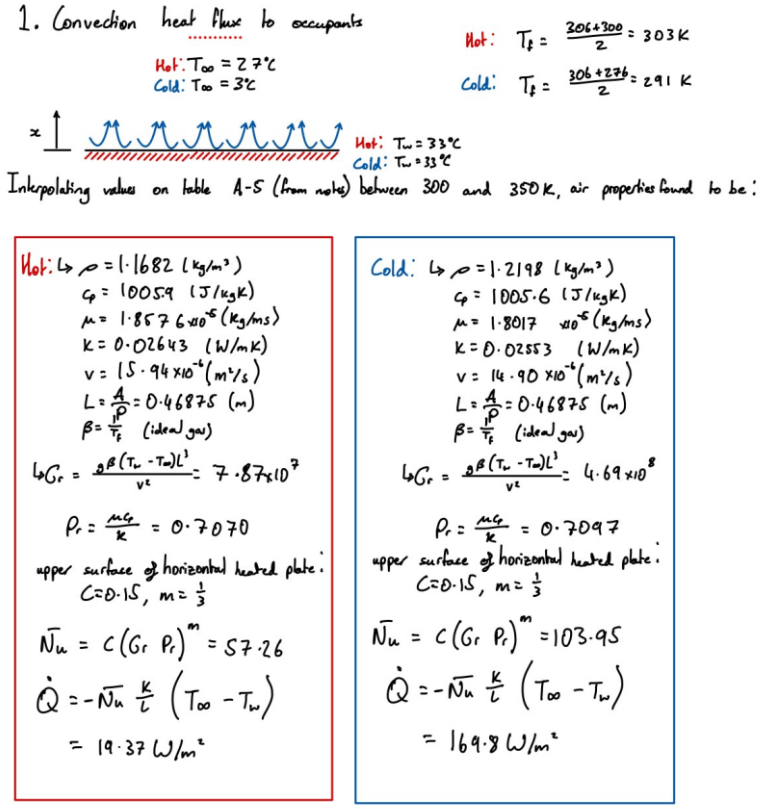
Radiative Heat Flux

Using the same assumptions and processes as described previously, the heat flux away from the occupants due to radiation was found to be -15 W/m^2 for the hot case and 227 W/m^2 for the cold case. The cold case assumes the occupants have entered the cold system and have no insulation, so the value for heat loss is extremely high and in reality, would result in the core temperature of the occupants reducing and cause a shift in temperatures. When discussing the values for the cold case later, this will be considered.



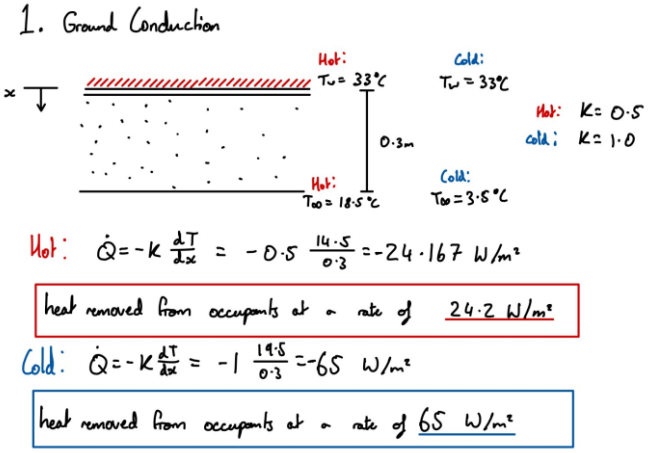
Convective Heat Flux

Following the same assumptions and processes as before the cases for hot and cold convective heat loss from the occupants' skin were found to be 19.37 W/m^2 in the hot case and 169.8 W/m^2 in the cold case. Again, following the assumption that the occupants were not insulated at all in the cold case.



Conductive Heat Flux

Analysing the ground conduction was much simpler than the convective and radiative analysis. The ground temperature values at 0.3m were taken from table [2] and the thermal conductivity for dry and moist soil (hot and cold cases respectively) found from [12] Using the model below and assumptions previously stated (the occupants being in perfect contact with the ground and the tent walls being thin), the heat flux in hot and cold cases removed from the occupants was calculated below as 24.2W/m^2 in the hot case and 65 W/m^2 in the cold case.



Discussion of Problem

Hot Case

For the hot case, the 3 losses of heat flux from the occupants can be found above, with a net heat flux of 28.6 W/m^2. When comparing this heat flux to the discomfort criteria, the internal heat storage of each occupant can be calculated as 29.6 W/m^2. This is higher than the 15.3 W/m^2 threshold and the occupants are uncomfortably hot. The majority of the heat flux to the occupants is from the radiation of the tent surface, something that should be considered in the solution proposal and aim to reduce the heat flux to the occupants by a minimum of 14.3 W/m^2.

Cold Case

For the cold case, the 3 losses of heat flux from the occupants can be found above, with a net heat flux of 461.8 W/m^2. When comparing this heat flux to the discomfort criteria, the internal heat storage of each occupant can be calculated as -403.6 W/m^2. This is significantly lower than the -15.3 W/m^2 threshold and the occupants are dangerously cold. The majority of the heat flux from the occupants is from convection and the radiation of the tent surface something that should be considered in the solution proposal and aim to reduce the heat flux from the occupants by a minimum of 388.3 W/m^2.

Proposing Solutions

As discussed above there were two scenarios to consider when proposing solutions and any design modifications should account for both. By insulating the tent more effectively to retain heat at night, during the day this would result in more heat being trapped in the tent and further discomfort. There was also the lightweight design that may be compromised if alterations were made to the tent. Fins and heat exchangers were considered to cool the surface of the tent but due to the additional weight of fins and the power source requirement (and the fact the tent would be unable to fold up for hiking and its intended purpose) of a heat exchanger, both of these solutions were rejected. From these requirements and limitations of design, other solutions were required.

Hot Case

For the hot case the key focus was to reduce the radiation from the tent surface to the occupants. This would be achieved by cooling down the surface of the tent that was seen to be 42 degrees C.

Passive Strategies

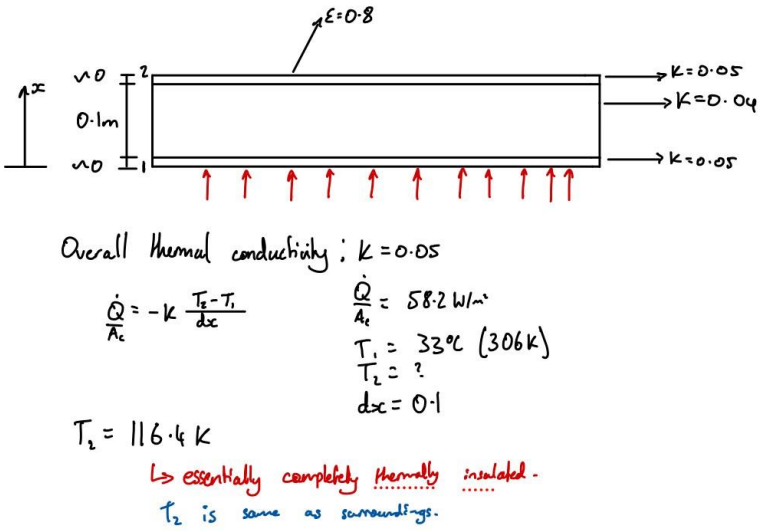
More ventilation in the tent: adding zipped vents along the sides of the tent would allow some airflow through the tent so instead of stagnant air (and a low heat transfer coefficient due). This would cool the air and assuming the air flow inside the tent to be half the wind speed of the outside air, a new Re can be calculated and using the same method of calculation as before, a new thermal conductivity can be calculated as 2.29 which corresponds to a new heat flux of 34.3 W/m^2 using Newton's law of cooling rather than the original value of 18.9 W/m^2 (an increase of 15.4 W/m^2) calculated in the convective calculations above. This modification alone would sufficiently remove energy from the system for it to be considered thermally comfortable (more than the suggested 14.3 W/m^2). Furthermore, using zipped ventilation would negate the effect these modifications would have at night.

Active Strategies

Solar outdoor fan: a solar powered fan to increase the wind speed over the outside of the tent surface would remove heat from the surface at a faster rate and cool the surface, in turn reducing radiation to the occupants. For the hot day there is an assumption of high intensity sun (900 W/m^2) using a solar panel of 20% efficiency [13] would harness (180 W/m^2). Whilst this solution could be analysed further, the consideration for weight and size of the fan would make it a less effective option.

Cold Case

As previously discussed, in the cold case the biggest issues were the losses due to lack of insulation of the occupants. The tent could be modified to retain more heat and even have a reflective coating; however, this would result in worsening of the hot case by making the tent retain more internal heat. There was also the weight consideration that limits the thickness of material to improve insulation. The most appropriate additional insulation would be around the occupants specifically in the form of a well-insulated sleeping bag. Below is a diagram of a proposed design for a sleeping bag with key material properties.



Conclusion

Whilst this report made large modelling assumptions and simplifications, the general process to find the heat flux experienced by a camper showed clear improvements needed to be made to improve thermal comfort. The best solution for a camper's thermal comfort is to consider the cases themselves and consider not pitching a tent in the direct sun or bring adequate insulation in the form of a sleeping bag with them. For this reason, the three suggestions for products that an outdoor equipment manufacturer may produce are: modifying tents with zipped and netted vents to allow stagnant air to be equalised by outside air and help to cool the surface of the tent; using a cotton filling and a polyester coating to thermally insulate the occupants of the tent; and a survivors' guide suggesting sensible considerations when camping such as pitching a tent in the shade or bringing a well-insulated sleeping bag and mat that create more barriers between the campers and the cold surroundings.

References

[1] Most popular camping location in the UK

[2] ASHRAE

[3] Ultimate outdoors website

[4] emissivity of polyester

[5] absorptivity of polyester

[6] PMC human skin temp: 33

[7] Emissivity of human skin

[8] Met office weather

[9] Soil info

[10] T hot inside air temp

[11] T cold inside air temp

[12] Thermal Conductivity eng toolbox

[13] Most Efficient Solar Panels In The UK For 2025 Ranked - iHeat